ST-Mamba: Spatio-Temporal Synergistic Model for Remote Sensing Change Detection

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***Abstract*— The advancement of remote sensing and deep learning has spurred interest in high-resolution image change detection (CD). However, pseudo-changes in multitemporal images, due to complex scenes and variable imaging conditions, often lead to significant misdetection in current methods. To address this problem, we propose a new CD framework: spatio-temporal (ST)-Mamba, which consists of three key components. First, a Mamba-based feature extraction module (MFEM) is designed as the encoder to extract essential features from multitemporal images by leveraging Mamba's capability to capture inherent information in long data sequences. Second, a ST synergistic module (STSM) is developed to unify the background features of multitemporal feature maps into a common domain by employing the state-space model (SSM) for ST modeling. Finally, an ST fusion module (STFM) is created to guide the fusion of image features at different scales and across channels by utilizing a feature map of the unified background features. The experimental results on five widely used CD datasets show significant improvements over current state-of-the-art (SOTA) methods.**

***Index Terms*— Change detection (CD), mamba, remote sensing image, spatio-temporal (ST) relationship, state-space model (SSM).**

I. Introduction

THE process of detecting variations by monitoring the condition of an object or phenomenon over time is known as change detection (CD) [1]. This technique is essential in various applications, including disaster response [2], agricultural management [3], urban planning [4], and environmental monitoring [5]. The complexity of real-world scenarios and the diversity of change objects that need to be recognized present significant challenges for CD. Different CD tasks focus on various change objects, such as roads, buildings, farmland, or multiple types of objects. Meanwhile, multitemporal remote sensing images contain both actual changes and pseudo-changes, the latter of which can distort CD results and lead to substantial false detections. These pseudo-changes arise from complex scenes and variable imaging conditions, primarily due to changes in objects that are not relevant to the detection task. For instance, in the architectural CD task, due to seasonal changes caused by withering vegetation [6], snow cover [7], or the appearance of other building-like objects. Differences in imaging conditions primarily arise from the varying shooting conditions encountered when capturing multitemporal images. These variations can include differences in illumination during the shoot [8], discrepancies in shooting angles or misalignment of shooting areas [9], and variations in resolution [10]. These changes are widely present in remote sensing images from the real world. The multitemporal images, which exhibit both complicated and varied actual changes as well as pseudo-changes, make it challenging for traditional CD techniques based on algebra [11], transformations [12], and classification [13].

The efficiency and accuracy of CD have been greatly increased by the introduction of novel models and methodologies, particularly with the emergence of deep learning techniques. Small-scale models were the main focus of early research [14], [15], [16], but the subsequent rise of massive datasets has led to the development of deeper models [17], [18], [19]. With the advent of fully convolutional networks (FCNs) [20], convolutional neural network (CNN)-based techniques have become prevalent in CD [21]. Nevertheless, CNNs have limitations when dealing with complex scenes [22], and it is difficult for CNN-based methods to capture the contextual information of the input images due to the fixed receptive field of the convolutional kernel. The Visual Transformer [22] addresses this issue by employing a self-attention mechanism, which effectively extracts global features from images while incorporating contextual information from multitemporal images. However, it still faces challenges related to computational costs when applied to large-scale datasets [23], [24], which limits the further development of the technique.

An alternative to the Transformer that excels in analyzing lengthy sequential data and offers the potential for linear scaling is the Mamba model [25]. Currently, the Mamba architecture has been applied to image data [26] and has been extended to the CD domain [27], [28], [29], where it demonstrates extremely advanced performance. However,

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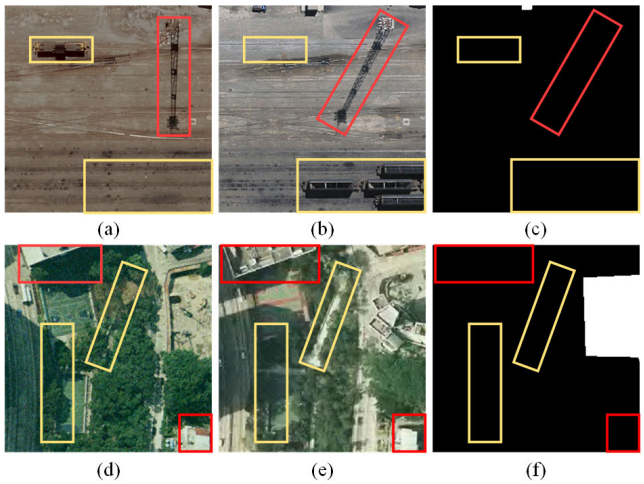


Fig. 1. Yellow boxes highlight the pseudo-changes resulting from scene complexity, while the red boxes indicate pseudo-changes caused by variations in imaging conditions. (a) and (d) $T1$ images. (b) and (e) $T2$ images. (c) and (f) Ground truth.

these methods mainly focus on feature extraction for multitemporal images while neglecting the impact of pseudo-changes on the detector. As shown in Fig. 1, there are significant differences in illumination between the images captured before and after the time phase. Specifically, in Fig. 1(a)–(c), the shadows of the pylons in the red box introduce substantial pseudo-changes due to the differing times of capture, and the long vehicles in the yellow box are easily misidentified due to their similarity to buildings. In Fig. 1(d)–(f), the trees within the yellow frame display significant changes attributed to seasonal variations, which obscure the areas that remain unchanged. Conversely, the building in the red frame appears altered due to differences in shooting angles, despite no actual changes occurring. These pseudo-changes can severely hinder effective changing feature extraction, resulting in substantial misdetection.

To improve this, spatio-temporal mamba (ST-Mamba) is proposed in this article, which is an effective exploration of the CD framework. It makes full use of the Mamba property to synergize multitemporal features while significantly reducing the impact of pseudo-changes on detection results. Specifically, we build the mamba-based feature extraction module (MFEM) to extract image features more efficiently and with reduced computational costs. Next, the ST synergistic module (STSM) is developed. It takes the initially extracted feature pairs as inputs and employs a state-space model (SSM), allowing multitemporal feature maps to synergize in ST modeling for a comprehensive understanding of background features. These complete background features are then applied to the single temporal feature map to unify the background features of multitemporal images into a unified domain, effectively diminishing the influence of pseudo-changes on feature extraction. However, the unification of background features is accompanied by the blending of changing features from different time phases, which results in the leakage detection of certain change objects. To address this problem, on the one hand, STSM enhances the changing features of the current time by employing an adaptively improved attentional mechanism. On the other hand, the ST fusion module (STFM) decoder is designed to guide the fusion of multiscale and channel features, effectively minimizing the loss of change objects while reducing the impact of pseudo-changes. To summarize, we make the following contributions.

1) We propose a novel CD network, ST-Mamba, which employs a new CD framework. It enables the synergistic effects of multitemporal features to greatly reduce the impact of pseudo-changes on feature extraction while effectively extracting the changing features.
2) The MFEM is designed to fully extract multitemporal features across all scales while effectively reducing computational complexity.
3) The ST synergistic module is designed to effectively utilize the SSM to synergize multitemporal features for ST modeling. This approach seeks to unify background features, thereby reducing the impact of pseudo-changes.
4) The STFM is proposed, which utilizes unified background features to guide the fusion of multiscale and cross-channel feature data, effectively ensuring the accurate detection of changing objects.

The remainder of this article is organized as follows. Section II reviews the related work. Section III introduces the details of ST-Mamba. Section IV presents the experimental results and discussion. Section V concludes our work.

II. Related Works

In this section, we focus on reviewing deep learning-based CD methods for remote sensing images.

A. CNN-Based CD Methods

CNN-based CD techniques are crucial in remote sensing image analysis, effectively extracting local features through hierarchical structures. Daudt et al. [20] first introduced FCNs for CD, proposing the FC-Siam-Conc model based on Siamese networks. These networks compare two temporal images by sharing weights, which reduces similarity distances and network parameters. Subsequently, Zhang et al. [21] enhanced multilevel fine-grained detection by extracting feature change maps at different phases. Shi et al. [17] incorporated deep supervision into the Siamese network decoder to improve label-level CD. Zhao et al. [30] proposed a dual-stream framework with semantic guidance to address challenges in bitemporal feature fusion. With the development of attention mechanisms, networks have increasingly utilized them to expand receptive fields and improve performance. Fang et al. [31] proposed a densely connected twin architecture with channel attention to enhance bitemporal feature interaction. Han et al. [32] integrated self-attention and hierarchical convolutions to improve global and local feature representation. Chen and Shi [6] combined spatial and channel attention mechanisms to enhance the network's ability to identify change regions from multiple perspectives. Zhou et al. [33] integrated twin networks with multihead self-attention to reduce imaging differences between multitemporal images.

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Diakogiannis et al. [34] incorporated dilated convolutions and pyramid pooling into a UNet framework for multiscale feature extraction. Gu et al. [35] focused on leveraging multiscale discrepancies between bitemporal images to highlight fine-grained details in changing regions.

Currently, CNN-based CD methods have significantly matured and are particularly good at extracting local features. However, CNN architectures make it difficult to capture long-range dependencies due to their inherent local receptive field properties [36]. This makes it difficult to accurately detect the change in regions in the face of CD tasks where actual changes are sparse and pseudo-changes are complex.

B. Transformer-Based CD Methods

The Transformer model has gained popularity in computer vision due to its effectiveness in modeling long-range dependencies [37], thereby overcoming the limitations of CNNs in capturing global spatial relationships. The Vision Transformer (ViT) [38] introduced Transformer networks to image analysis, capturing long-range dependencies that CNNs could not achieve. Chen et al. [22] proposed a method to transform multitemporal images into semantic tokens, facilitating the modeling of spatial–temporal relationships within a token-based framework and significantly enhancing the CD process. Bandara and Patel [39] introduced a twin network with a pure Transformer design, combining a hierarchical encoder and a multilayer perceptron decoder to capture multiscale details while reducing CNN reliance. Li et al. [40] proposed a hybrid model that integrates the global contextual modeling capabilities of Transformers with CNN blocks within a UNet framework. This approach eliminates redundant information in the UNet architecture and enhances feature quality through a difference enhancement module, thereby ensuring more accurate detection. Zhang et al. [41] used the Swin Transformer [23] as a backbone in a twin U-Net structure for multilevel feature extraction, recovering detailed change information and improving performance with a channel attention mechanism.

Transformer-based methods enhance the accuracy of CD tasks but face significant computational challenges when processing large-scale, high-resolution images. To tackle this issue, researchers developed the Mamba architecture [25], which retains the nonlocal modeling capabilities of the Transformer while ensuring linear computational scaling. This advancement substantially alleviates the burden of large-scale image processing while preserving global contextual information essential for complex prediction tasks.

C. Mamba-Based CD Methods

Recently, the SSM Mamba has been seen as a promising alternative to Transformer due to its linear computational complexity when dealing with long sequences [42]. It has shown potential for dealing with long-term dependencies in vision tasks. The field of CD is also beginning to adopt the Mamba architecture. Our study will compare with these works. Chen et al. [27] first applied Mamba to the field of CD by combining VMamba [26] and twin-shared weight structure as an encoder and devised three different traversals in the decoder to model ST relationships among multitemporal images. Huang et al. [43] incorporated Mamba into the land cover CD by using a Vmamba-based twin network and different token modeling methods to strengthen the correlation of changing features across dimensions. Chen et al. [28] integrated the diverse observation angles of remotely sensed imagery into VMamba and introduced four oblique scanning methods to capture more comprehensive spatial information. Zhang et al. [29] designed a new Mamba module in combination with Vim [44] and combined the mutual bootstrap difference results of multitemporal phase images and multiscale features in the decoding process, which significantly improved the network performance.

However, current Mamba-based CD networks struggle with complex and varied pseudo-changes, as Mamba's state accumulation can significantly impact the network's overall perception, leading to substantial detection errors. Our proposed ST-Mamba network effectively solves this problem by synergizing multitemporal images, unifying the background features into a unified domain, and then using the unified features to guide the fusion of multiscale and multichannel features. Experiments indicate that our approach markedly surpasses other Mamba-based CD networks across five datasets, with visualizations confirming this improvement.

III. Methodology

A. Preliminaries

In the landscape of deep learning research, SSMs have emerged as a powerful paradigm for modeling dynamic systems. These models, which adhere to the linear time-invariant (LTI) property, are particularly adept at mapping input sequences $x(t) \in \mathbb{R}$ to corresponding output sequences $y(t) \in \mathbb{R}$ through the mediation of a hidden state $h(t) \in \mathbb{R}^N$, where N is the size of the hidden state. SSMs are deeply rooted in control theory and are characterized by a suite of linear ordinary differential equations (ODEs) that describe the hidden state's temporal evolution

$$h'(t) = Ah(t) + Bx(t) \tag{1}$$

$$y(t) = Ch(t) \tag{2}$$

where $A \in \mathbb{C}^{N\times N}$, $B \in \mathbb{R}^{N\times 1}$, and $C \in \mathbb{R}^{1\times N}$ are the system matrices and $h(t)$ denotes the hidden state vector at time t. Subsequently, to incorporate continuous-time models into deep learning, a time interval $\mathbf{\Delta}$ is often used to discretize the continuous matrices A and B through a zero-order hold (ZOH) [45] method, yielding discrete matrices $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$

$$\bar{\mathbf{A}} = \exp(\mathbf{\Delta A}) \tag{3}$$

$$\bar{\mathbf{B}} = (\mathbf{\Delta A})^{-1}(\exp(\mathbf{\Delta A}) - \mathbf{I})(\mathbf{\Delta B}). \tag{4}$$

Following the ZOH discretization of matrices A and B into discrete matrices $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$, (1) and (2) are then represented in their discrete forms:

$$h(t) = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \tag{5}$$

$$y(t) = \mathbf{C}h_t. \tag{6}$$

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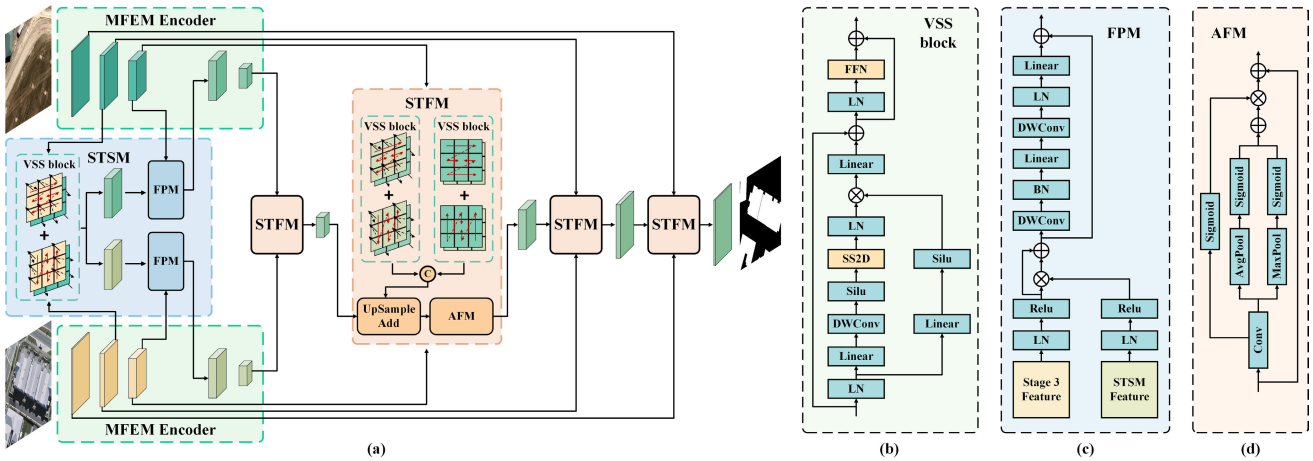


Fig. 2. Illustration of our method. (a) Overview of the ST-mamba framework. Given a pre-change and post-change image, the initial extraction of features is performed by the MFEM. These features are then fed into the STSM, where they are unified into a unified domain through a synergistic effect. Finally, the unified features are fed into a decoder consisting of STFM, which integrates multiscale and multichannel features to generate the final change map. (b) Schematic of the structure of the VSS blocks. (c) Structure schematic of the FPM in the STSM. (d) Structure schematic of the AFM in the STFM.

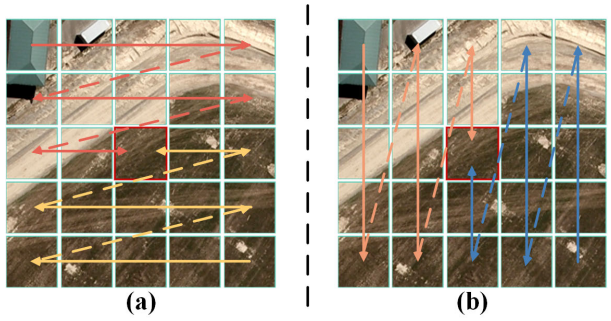


Fig. 3. Illustration of the cross-scan mechanism [26]. In this example, (a) horizontal cross-scan and (b) vertical cross-scan used to capture global contextual information.

The final output can be obtained directly through full convolution computation.

Mamba enhances SSMs by introducing data-dependent learnable parameters $\mathbf{\Delta}$, $\bar{\mathbf{B}}$, and $\mathbf{C}$, which dynamically adjust to different inputs, addressing the issue of constant parameters in traditional SSMs. This approach, coupled with a hardware-aware algorithm for GPU optimization, enables efficient processing of long sequences. Mamba's architecture filters out irrelevant information and recalls relevant details, offering fast training and inference with memory and computational costs that scale linearly with sequence length.

B. ST-Mamba Framework

An overview of the proposed ST-Mamba framework is shown in Fig. 2(a). The ST-Mamba framework comprises three main components: the MFEM, the STSM, and the STFM. Specifically, the MFEM consists of four layers of feature encoders, each incorporating visual state-space (VSS) blocks and downsampling modules. During the processing of feature maps, the features of pre-change and post-change images are initially extracted by the first three layers of feature encoders, respectively. The feature maps obtained from the second layer encoder are then input into the STSM, where the ST information of the multitemporal feature maps is utilized for synergistic modeling. This process also fuses the feature maps output from the third layer encoder to unify the background features of the pre-change and post-change feature maps into a unified domain. Subsequently, the two feature maps output from the STSM are fed into the fourth layer encoder of their respective branches to further extract features while obtaining feature maps at more scales. Finally, a four-layer feature decoder is constructed with the STFM as the primary component. This decoder gradually fuses multiscale and multichannel features, utilizing a unified background feature map to guide the generation of the change mask. This approach aims to prevent the loss of actual change objects while minimizing the influence of pseudo-changes on CD.

In this article, we call the encoder of each layer $E_i^{T_1}$ and $E_i^{T_2}, i = 1, 2, 3, 4$, where $T1$ represents the branch of the preprocessed temporal image, $T2$ represents the branch of the postprocessed temporal image, and subscript i denotes the encoder of the ith layer. Call the decoder of each layer as D_i, $i = 1, 2, 3, 4$, and subscript i denotes the ith layer decoder. In Sections III-C–III-E, we will detail each of the three components of ST-Mamba.

C. Mamba-Based Feature Extraction Module

MFEM serves as the foundational module for constructing a four-layer decoder. Each layer of the decoder comprises a downsampling module and an MEFM, which consists of VSS modules with varying numbers of layers. The downsampling multiplier is set to 2 for all layers, except for the first layer of the decoder, which utilizes a downsampling multiplier of 4, in accordance with the configuration in VMamba. In a single VSS block, as shown in Fig. 2(b), the input first passes through a layer normalization (LN) layer and a linear embedding layer. The output is then divided into two streams of information. One stream is processed through a depth-wise convolution (DWConv) layer, typically employing a 3×3 convolution. This stream is subsequently fed into the core SS2D module after passing through the SiLU activation function [46]. The

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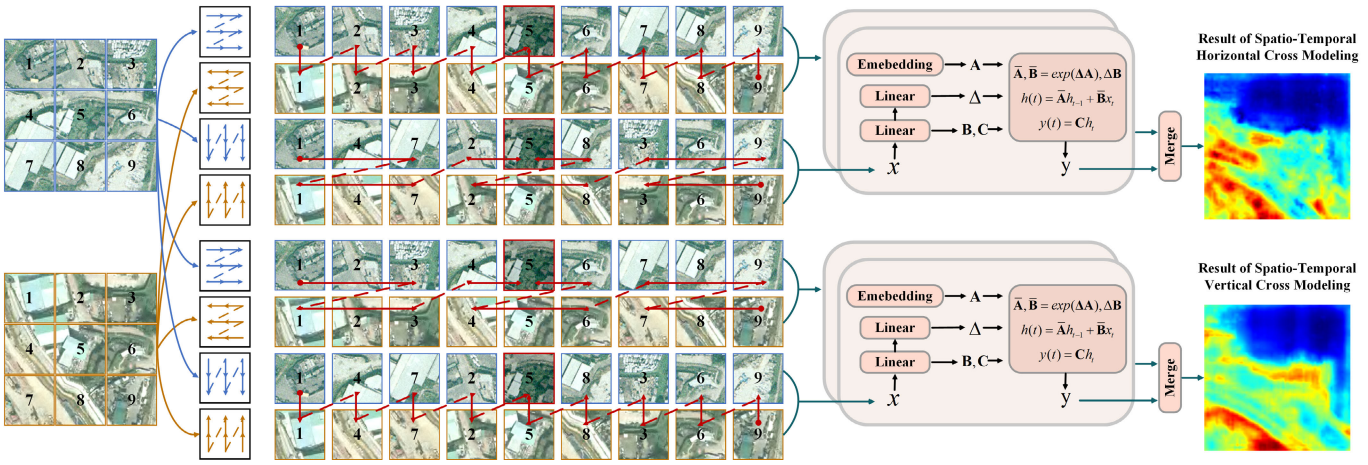


Fig. 4. Following two ST modeling processes in STSM are demonstrated: STHCM and STVCM. The blue color in the figure represents the feature information of pre-change images, and the yellow color represents the feature information of post-change images. In the schematic with blue and yellow arrows, the horizontal direction indicates horizontal scanning, and the vertical direction indicates vertical scanning. The red arrow represents the scanning direction used in the ST modeling process.

output from the SS2D module is then passed through the LN layer and is combined with the output of the other stream, which has also been activated by SiLU. Finally, the information stream is processed by a residual network that includes the LN layer and the feed-forward neural (FFN) layer, producing the final output of the VSS block. The data are processed through the layers as described, with the exception of the fourth encoder layer, which takes the STSM output as its input

$$E_{i+1}^{T_x} = E_v\left(E_{\mathrm{DS}}\left(E_i^{T_x}\right)\right) \tag{7}$$

where E_{DS} is the downsampling module for each layer; E_v is the VSS block for each layer; $x = 1, 2$; and $i = 1, 2$.

Compared to the VSS block in VMamba [26] and MambaCD [27], MFEM includes further modifications to the VSS block, enhancing its suitability for the CD task, in addition to the implementation of FFN networks. In specific applications, the current Mamba-based CD networks exhibit a high susceptibility to overfitting. To solve this issue, the number of VSS blocks in MFEM has been adjusted to four for the third layer, whereas MambaCD and RS-Mamba [28] utilize 15 and 9 blocks, respectively (RS-Mamba offers 1.5 times more SS2D scanning directions than MambaCD.) Furthermore, the cross-scanning mechanism for SS2D in VMamba has been streamlined in MFEM. The horizontal and vertical cross-scans are illustrated in Fig. 3. In Vmamba, the output value of SS2D is the superposition of the results from these two scans. In MFEM, the process is modified so that only horizontal scanning is conducted in the first VSS block, while only vertical scanning is performed in the subsequent VSS block, and this pattern continues in pairs of two blocks. The number of VSS blocks in the four-layer encoder is [2, 2, 4, 2].

D. ST Synergistic Module

The presence of pseudo-changes in multitemporal feature maps significantly adversely affects current CD algorithms. In particular, Mamba-based CD methods are susceptible to erroneous feature accumulation due to pseudo-changes, resulting in substantial omissions and misdetections. To mitigate the impact of pseudo-changes on the CD task, the STSM employs SSMs to facilitate synergistic effects among multitemporal feature maps in ST modeling. This approach unifies the background features of the pre-change and post-change feature maps into a unified domain.

Specifically, as shown in Fig. 2(a), STSM takes the feature maps output from $E_2^{T_1}$ and $E_2^{T_2}$ as inputs to this module. The structure of the VSS block employed in this module is similar to that in MFEM, with the distinction that both pre-change and post-change feature maps are utilized synergistically for feature extraction. In STSM, two ST modeling approaches are developed: ST horizontal cross modeling (STHCM) and ST vertical cross modeling (STVCM), as shown in Fig. 4. In the STHCM process, horizontal scanning effectively extracts unified background features in the horizontal direction. However, it is prone to misidentifying relevant features of changing objects as background features. Since the background features of multitemporal images are more complex than those of a single image, a more integrated global modeling process is required. To address this issue, a cross-scan modeling approach is employed. In this approach, vertical scanning compensates for features that are subject to change. The features obtained from both scanning directions are then merged to produce the final output of the STHCM process. Similarly, in the STVCM process, vertical scanning emphasizes the vertical unified background features for ST modeling, while horizontal scanning compensates for changing features. The results from both scans are merged to yield the outcomes of the STVCM process.

In STSM, the feature maps generated from $E_2^{T_1}$ and $E_2^{T_2}$ are processed together with STHCM and STVCM. These processed feature maps are then input into the feature processing module (FPM) alongside the outputs of $E_3^{T_1}$ and $E_3^{T_2}$, respectively. The FPM utilizes the extracted uniform background features to modify the output features of $E_2^{T_1}$ and $E_2^{T_2}$ to unify the background features to the same domain. This approach aims to minimize the impact of pseudo-changes

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TABLE I

COMPARISON OF ST-MAMBA WITH RESPECT TO SOTA CD METHODS. $F1$ DENOTES THE $F1$ METRIC, IOU DENOTES THE INTERSECTION-OVER-UNION METRIC, AND OA DENOTES THE OVERALL PIXEL ACCURACY. THE BEST RESULT IS INDICATED IN BOLD AND THE SECOND-BEST RESULT IS UNDERLINED. OUR METHOD OUTPERFORMS EXISTING METHODS FOR ALL DATASETS

	SYSU			WHU-CD			LEVIR-CD			CDD			LEVIR-CD+		
Method	OA	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA	F1	IoU
CNN-based Methods															
FC-EF [20]	88.69	75.81	61.04	98.87	84.89	73.74	98.89	88.90	80.03	97.43	88.55	79.45	98.08	76.41	61.82
FC-Siam-Diff [20]	88.65	75.79	61.01	99.13	87.67	78.04	98.96	89.69	81.31	98.07	91.74	84.74	98.34	79.23	65.61
FC-Siam-Conc [20]	88.05	75.18	60.23	98.94	85.83	75.18	98.98	89.89	81.64	97.75	90.14	82.05	98.33	79.09	65.42
DSIFN [21]	89.59	78.80	65.02	99.20	89.73	81.37	98.81	88.11	78.75	99.04	95.89	92.12	98.41	80.06	66.76
SNUNet [31]	87.49	73.14	57.66	99.10	87.70	78.09	98.85	88.59	79.51	99.06	96.17	92.61	98.26	78.56	64.70
Transformer-based Methods															
BIT [22]	90.60	79.62	66.14	99.26	90.01	81.84	99.01	90.03	81.87	98.95	95.58	91.53	98.43	80.95	68.00
SwinsUNet [41]	91.27	80.16	66.89	99.24	91.35	84.08	98.77	87.77	78.21	97.92	94.36	89.32	98.22	78.31	64.35
Mamba-based Methods															
MambaCD [27]	91.31	80.39	67.22	99.22	91.49	86.41	99.08	90.54	83.40	99.10	96.19	92.67	98.57	82.68	70.47
RS-Mamba [28]	90.59	78.99	65.28	99.27	91.97	85.14	99.08	90.82	83.18	98.99	96.06	92.42	98.50	81.07	68.16
CDMamba [29]	90.33	79.07	65.38	99.20	91.10	83.65	99.06	90.75	83.07	98.67	94.38	89.36	98.51	81.77	69.16
ST-Mamba (ours)	**91.95**	**83.39**	**71.50**	**99.52**	**94.72**	**90.15**	**99.19**	**91.92**	**85.05**	**99.42**	**97.76**	**95.64**	**98.73**	**84.01**	**72.44**

on feature extraction while simultaneously preserving the essential changing features. The processing of the STSM can be expressed by the following equation:

$$S_{\text{out}}^{T_i} = \text{FPM}\left(E_3^{T_i}, S_{\text{st}}\left(E_2^{T_1}, E_2^{T_2}\right)\right) \tag{8}$$

where $S_{\text{out}}^{T_i}$ denotes the output result of STSM; FPM$(\cdot)$ denotes the feature processing module; and $S_{\text{st}}(\cdot)$ denotes two ST modeling processes applied to the VSS block in STSM, $i = 1, 2$. Finally, the output results of STSM are input into $E_4^{T_1}$ and $E_4^{T_2}$, and the results obtained are involved in the decoder process.

E. ST Fusion Module

As shown in Fig. 2(a), the decoder, which primarily comprises the STFM, is tasked with summarizing the multiscale ST feature information extracted during the encoding stage to produce precise predictions of change regions. In total, there are four layers of decoders, each consisting of an STFM module along with an upsampling module.

First, the feature maps obtained from the $E_i^{T_1}$ and $E_i^{T_2}$ layers during the encoding stage are input into the corresponding decoding D_i layer. Next, two ST modeling approaches are employed to fuse the multiscale feature maps. One of these approaches is identical to the STHCM described in Section III-D. The other approach involves concatenating the feature maps output from $E_i^{T_1}$ and $E_i^{T_2}$, which are then processed by the VSS block. The structure of the VSS block is similar to that of the MFEM outlined in Section III-C, with the key difference being that SS2D utilizes a cross-scanning method, which combines the results of horizontal and vertical scans. The outputs from these two ST modeling approaches are concatenated and subsequently input into the adaptive fusion module (AFM) after being summed with the upsampling results from the previous layer of the decoder. The structure of the AFM is shown in Fig. 2(d), and it is designed to enhance the fusion of feature information across channels, thereby mitigating the loss of features associated with changing objects. Notably, the input to the D_4 layer is derived from the $E_4^{T_1}$ and $E_4^{T_2}$ layers, which consist of pairs of feature maps after the background information has been unified. Consequently, the output of the D_4 layer carries significant weight as the initial change map, guiding the subsequent feature fusion process. The decoding process of STFM can be expressed by the following equation:

$$D_i = \text{AFM}\Big(\text{UpAdd}\Big(\text{concat}\Big(S_{\text{sth}}\Big(E_i^{T_1}, E_i^{T_2}\Big), S_c\Big(E_i^{T_1}, E_i^{T_2}\Big)\Big), D_{i-1}\Big)\Big) \tag{9}$$

where AFM$(\cdot)$ refers to processing with the adaptive fusion module, UpAdd$(\cdot)$ denotes processing with the upsampling summation operation, $S_{\text{sth}}(\cdot)$ indicates processing with the VSS block of the STHCM method, and $S_c(\cdot)$ represents processing with the VSS block of the second concatenation modeling method in the STFM. Subsequently, the feature maps processed by the four-layer decoder are upsampled to match the size of the original image. The classifier then categorizes each pixel as either altered or unaltered based on the final feature maps, resulting in the final CD outcomes.

IV. EXPERIMENTS

A. Dataset Introduction

We compare our approach with current state-of-the-art (SOTA) CD methods on five widely used public datasets: the Sun Yat-sen University CD Dataset (SYSU) [17], Wuhan University Building CD Datasets (WHU-CD) [47], LEarning, VIsion and Remote sensing CD Datasets (LEVIR-CD) [6], LEVIR-CD+ [6], and CD Dataset (CDD) [7].

1) SYSU Dataset: The SYSU dataset [17] is a category-agnostic CD dataset comprising 20 000 pairs of 0.5-m/pixel resolution aerial images from Hong Kong,

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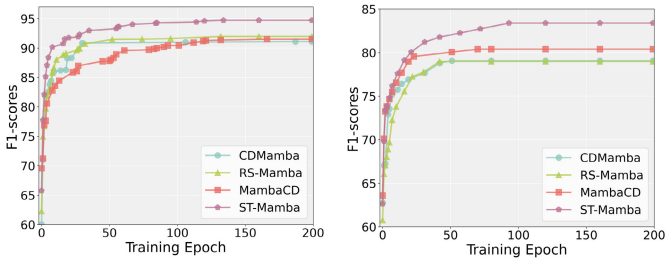


Fig. 5. Graphs show the changes in optimal $F1$-scores on the test set during the training process for each experimental method. The line graph maintains the best-result node every 40 iterations to ensure continuity, even if no superior results are obtained subsequently. (a) Results on the WHU-CD dataset. (b) Results on the SYSU dataset.

captured between 2007 and 2014. It includes complex change scenes such as urban construction, vegetation changes, and infrastructure development. The dataset is characterized by its complexity due to shadows and deviations in high-rise buildings. The data are divided into training, validation, and testing sets at a ratio of 6:2:2, with each set containing 256 × 256 patches [17].

2) WHU Dataset: The WHU-CD dataset [47] is a subset of the WHU Building dataset, focusing on building CD. It includes two aerial datasets from Christchurch, New Zealand, captured in April 2012 and 2016 with a spatial resolution of 0.3 m/pixel. The 2012 images cover 20.5 km^2 with 12 796 buildings, while the 2016 images show an increase to 16 077 buildings. The dataset is cropped into nonoverlapping 256 × 256 patches and split into training, validation, and testing sets at a ratio of 7:1:2 [48].

3) LEVIR-CD Dataset: The LEVIR-CD dataset [6] contains 637 pairs of very high-resolution (VHR) Google Earth images with a resolution of 0.5 m/pixel, spanning 5–14 years. It focuses on building-related changes and includes 31 333 annotated change instances. The images are cropped into 256 × 256 patches, either overlapping by 128 pixels or nonoverlapping, and divided into training, validation, and testing sets at a ratio of 7:1:2 [49].

4) LEVIR-CD+ Dataset: The LEVIR-CD+ dataset is an extension of LEVIR-CD, containing 985 pairs of VHR images at 0.5 m/pixel, each 1024 × 1024 pixels, captured between 2002 and 2020. It documents significant building changes over 5–14 years and includes approximately 80 000 building instances and 31 333 change instances. The images are split and cropped into patches for training, validation, and testing, following the same approach as LEVIR-CD.

5) CDD Dataset: The CDD dataset [7] consists of 11 pairs of season-varying Google Earth images capturing changes in buildings, roads, vehicles, and vegetation, excluding seasonal and brightness variations. It is challenging for CD algorithms. The high-resolution images are cropped into 256 × 256 patches, with 10 000 for training, 3000 for validation, and 3000 for testing, following the splits in [39].

B. Experimental Setup

We use the AdamW optimizer [50] with initial learning rate 1e^{-3} and weight decay 5e^{-3}. The batch size during training is kept at 16 and the model is trained with 240 epochs on each dataset. The parameters associated with Mamba are identical to those of the Base model in VMamba [26], with the exception that the number of layers in the VSS block is configured to [2, 2, 4, 2]. Data augmentation on the training set is performed by randomly flipping and exchanging the order of the bitemporal images. It is worth noting that neither the proposed method nor the Mamba-based CD method uses pretraining weights in the experimental setup to ensure the fairness of the experiments. Our loss function integrates binary cross-entropy loss with Dice coefficient loss. We use PyTorch [51] to implement the ST-Mamba and train it on an RTX 3090 GPU (24-GB memory). Following [52], we use three metrics for evaluation—$F1$-score, intersection over union (IoU), and overall accuracy (OA).

C. Performance Comparison

To verify the effectiveness of ST-Mamba in CD tasks, some classic or SOTA methods are selected for comparison in this section, including CNN-based methods (FC-EF [20], FC-Siam-Diff [20], FC-Siam-Conc [20], DSIFN [21], and SNUNet [31]), Transformer-based methods (BIT [22] and SwinsUnet [41]), and Mamba-based methods (MambaCD [27], RS-Mamba [28], and CDMamba [29]). For a fair comparison, all methods are trained under the same conditions based on the officially published PyTorch code.

1) Quantitative Results: Table I illustrates the overall performance of all compared methods on the SYSU, WHU-CD, LEVIR-CD, CDD, and LEVIR+-CD test sets in terms of values. The best result is indicated in bold and the second-best result is underlined. It is evident that our proposed ST-Mamba demonstrates superior performance compared to CNN-based methods, Transformer-based methods, and Mamba-based methods.

In particular, the Mamba-based method demonstrates superior performance compared to the Transformer- and CNN-based methods across the five datasets. The proposed ST-Mamba method demonstrates optimal performance across all three metrics: OA, $F1$, and IoU. In particular, on the SYSU dataset, ST-Mamba demonstrates an improvement of 3% in the $F1$-score and 4.28% in the IoU score compared to the suboptimal MambaCD method, utilizing the same training conditions without using pretraining weights. Meanwhile, RS-Mamba and CDMamba demonstrate significantly better performance than the CNN-based methods, although they perform slightly worse than the Transformer-based methods. On the WHU-CD dataset, ST-Mamba also demonstrates a notable enhancement in performance, exhibiting a 2.75% improvement in $F1$-scores and a 3.74% improvement in IoU scores compared to the suboptimal outcomes. This indicates that ST-Mamba performs better in both class-wide and class-specific CD tasks. Meanwhile, among the methods compared, the Mamba-based approaches demonstrate more consistent performance and surpass the CNN- and Transformer-based methods. Furthermore, the performance on the LEVIR-CD, CDD, and LEVIR+-CD datasets also demonstrates a notable enhancement, with 1.1%, 1.57%, and 1.33% improvements in $F1$-scores and 1.65%,

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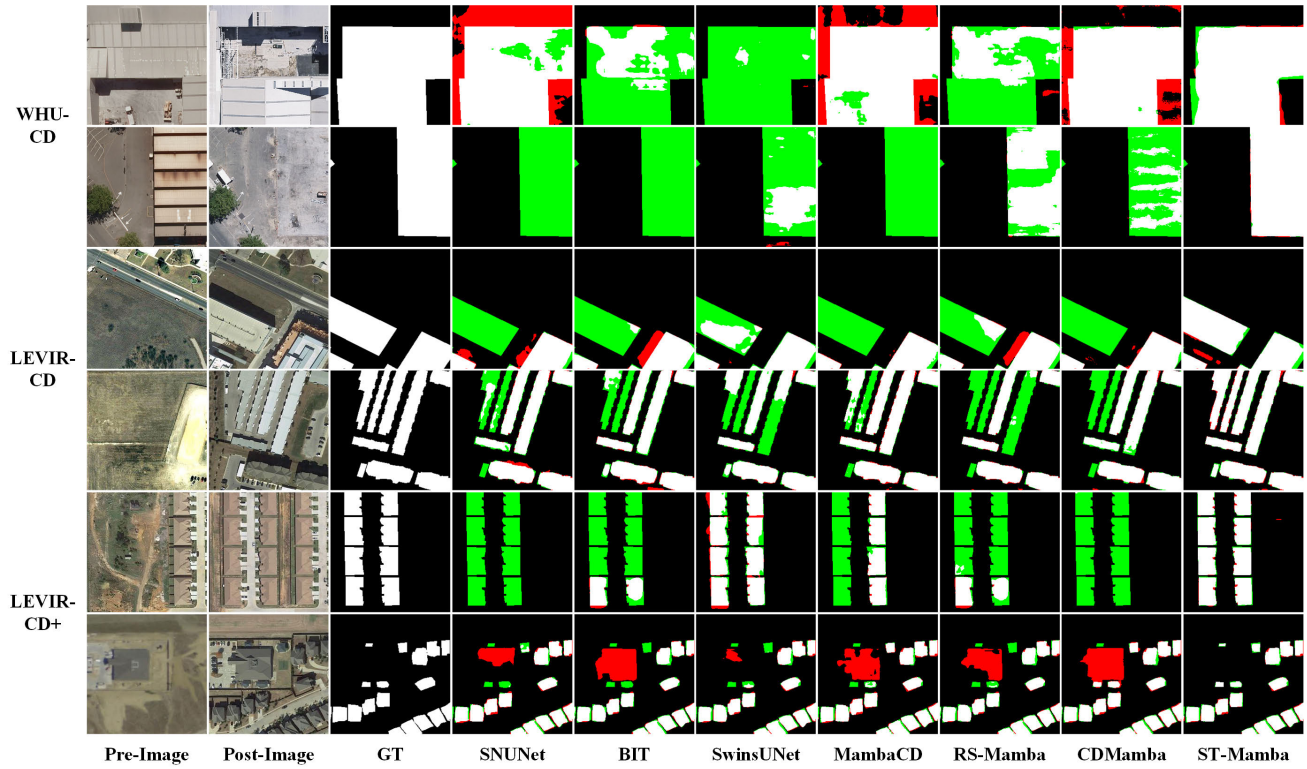


Fig. 6. Qualitative results on the WHU-CD, LEVIR-CD, and LEVIR-CD+ datasets. Pre-images and post-images are images captured before and after the change, respectively. GT denotes the ground truth. In the result plots for each method, white indicates true positives, black indicates true negatives, green indicates false negatives, and red indicates false positives.

2.97%, and 1.97% improvements in IoU scores, respectively, in comparison to the suboptimal outcomes. The results show that our proposed ST-Mamba exhibits superior performance in the context of multiple CD tasks and complex situations. Although the disparity between the Mamba-based methods across these three datasets is not as pronounced as in the first two datasets, the performance advantage over the CNN- and Transformer-based methods is considerably more significant.

In addition, ST-Mamba also reaches the best results faster during the training process. Fig. 5 shows the evolution of results on the test set for the optimal weight at each stage during training on the WHU-CD and SYSU datasets. It is clear from Fig. 5(a) that on the WHU-CD dataset, ST-Mamba achieves optimal results in the fewest number of training rounds. In contrast, almost all models are highly susceptible to overfitting on the SYSU dataset. This leads to a difference of more than 10% between the $F1$-scores of the current model on the SYSU and CDD datasets, even though the SYSU and CDD datasets belong to the same type of CD dataset and have similar data volumes. Compared to other Mamba-based CD networks, ST-Mamba excels in this regard, as shown in Fig. 5(b). While it achieves superior performance with the same number of training rounds, it effectively mitigates the issue of the dataset's high susceptibility to overfitting.

2) Qualitative Results: To further illustrate the validity of the proposed method, qualitative analyses are conducted on the SYSU, WHU-CD, LEVIR-CD, CDD, and LEVIR+-CD test sets. The accuracy of the tests is indicated by different colors: true positive (white), true negative (black), false positive (red), and false negative (green). For each dataset, two sets of visualization results are selected to demonstrate the effectiveness of the ST-Mamba improvement in comparison to the SOTA method. The WHU-CD, LEVIR-CD, and LEVIR+-CD datasets are specifically designed to detect changes in a particular class of objects, in this case, buildings, within multitemporal images. In contrast, the SYSU and CDD datasets require the detection of changes across all classes of objects in the multitemporal images. The visualization results of these five datasets will be analyzed next.

In Fig. 6, visualization results for the WHU-CD, LEVIR-CD, and LEVIR-CD+ datasets are shown. In building CD, other methods often misidentify entire change objects or misclassify change boundaries due to pseudo-changes. For example, in the WHU-CD dataset, variations in lighting and scene differences significantly impact other methods, and vehicles resembling buildings cause confusion in change region identification. In contrast, ST-Mamba exhibits minimal misdetection and omission, demonstrating high accuracy even in the presence of scene transformations and confounding differences. In the LEVIR-CD and LEVIR-CD+ datasets, the uneven distribution of buildings poses a challenge. Other methods frequently misidentify entire buildings, as seen in rows six and seven. In row 6, the bare ground's color closely matches the house, and the greenery in front resembles the original vegetation, causing other methods to miss multiple buildings. Although SwinUNet avoids misses, it is less effective than ST-Mamba in accurately delineating change boundaries. A similar issue occurs in row seven, where bare

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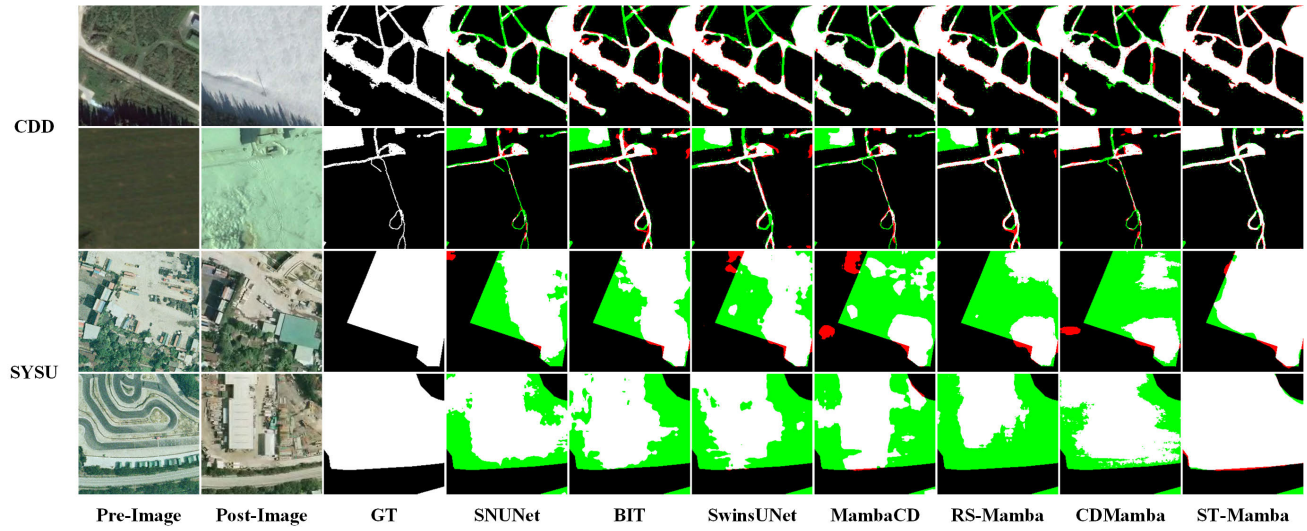


Fig. 7. Qualitative results on the CDD and SYSU datasets. Pre-images and post-images are images captured before and after the change, respectively. GT is the ground truth. In the result plots for each method, white indicates true positives, black indicates true negatives, green indicates false negatives, and red indicates false positives.

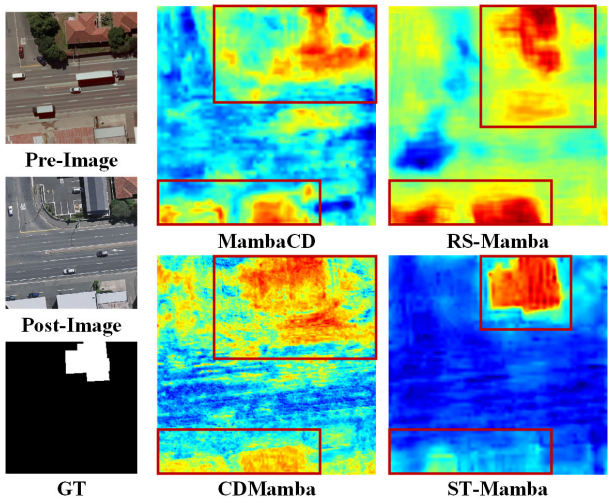


Fig. 8. Heatmap visualization of the feature maps for the WHU-CD dataset illustrates the probability of change for each pixel. Areas closer to red indicate a higher likelihood of change, while areas closer to blue suggest a greater probability of nonchange. The red boxes highlight the parts that are specifically compared in this article.

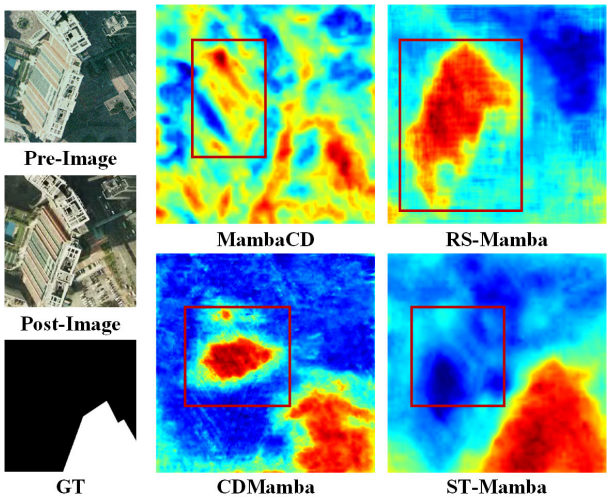


Fig. 9. Heatmap visualization of the feature maps for the SYSU dataset illustrates the probability of change for each pixel. Areas closer to red indicate a higher likelihood of change, while areas closer to blue suggest a greater probability of nonchange. The red boxes highlight the parts that are specifically compared in this article.

land's color resembles the building, leading to omissions and misdetections by other methods. ST-Mamba effectively addresses these issues.

The visualization results for the SYSU and CDD datasets are presented in Fig. 7. In the CDD dataset, the primary pseudo-changes arise from seasonal variations, necessitating the CD algorithm to identify obscured change objects, such as roads and houses, within dense vegetation and snow. In the examples shown in the first and second rows, all other methods produce detection errors for roads and some houses because these change objects are concealed by vegetation or snow. In contrast, ST-Mamba effectively addresses the pseudo-changes caused by seasonal variations by unifying the background information of the feature maps from both pre-change and post-change images into a unified domain, resulting in only minor misdetections of certain change boundaries. In the SYSU dataset, the main pseudo-changes stem from the complexity of the scene, characterized by variations in imaging angles and illumination. While other methods struggle to accurately identify change objects and boundaries, ST-Mamba successfully delineates the extent of the change objects and frames the area of the change region. However, ST-Mamba still faces challenges when dealing with cut building edges.

From the heat map, it is evident that ST-Mamba significantly attenuates the impact of pseudo-changes on CD results compared to other Mamba-based methods. For this comparison, we selected the WHU-CD dataset, which focuses on class-specific change objects, and the SYSU dataset, which encompasses all classes, as shown in Figs. 8 and 9. In Fig. 8, other Mamba-based methods are adversely affected by pseudo-changes, such as changes in house colors, variations

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TABLE II

FOLLOWING RESULTS DEMONSTRATE THE EFFICIENCY OF THE MODEL IN QUESTION. THE NUMBER OF GIGA FLOATING-POINT OPERATIONS PER SECOND (GFLOPS) AND THE TRAINING TIME (TIME) FOR A SINGLE EPOCH ON SYSU ARE REPORTED

Model	GFLOPs	Time (Min)
DSIFN	58.17	7.27
SNUNet	82.30	5.97
BIT	67.80	10.15
SwinUnet	45.71	10.90
MambaCD	107.67	13.32
RS-Mamba	57.95	9.68
CDMamba	67.18	15.52
ST-Mamba (ours)	98.02	7.12

TABLE III

ABLATION STUDY OF DIFFERENT COMPONENTS ON THE SYSU DATASET

Model	OA	F1	IoU
Baseline	91.10	80.63	67.55
Baseline + STFM	91.33	81.34	68.55
Baseline + ST-structure	91.63	81.66	69.00
Baseline + STSM	91.64	81.70	69.07
Baseline + ST-structure + STFM	91.90	82.57	70.31
Baseline + STSM + STFM	91.95	83.39	71.50

in lighting, and vehicles that resemble houses. This results in inaccurate detection of actual changes and the erroneous identification of pseudo-changed areas as genuine changes. In contrast, ST-Mamba remains largely unaffected by these issues, with its detected change regions closely aligning with the GT image. Similarly, in Fig. 9, other methods are influenced by variations in illumination and shooting angles, while ST-Mamba effectively mitigates these effects, with detected changes closely corresponding to the GT images.

3) Model Efficiency: To further validate the efficiency of the proposed model, Table II shows the GFLOPs and the time taken to train one epoch on the SYSU dataset (Time). Compared to other methods, the GFLOPs of our ST-Mamba are of a similar order of magnitude, while the training speed of ST-Mamba is superior. In our actual training process, when the batch size is set to 16, both MambaCD and CDMamba are out of memory, with 84.7% of the memory occupied by RS-Mamba and 72.19% of the memory occupied by ST-Mamba. This fully demonstrates that our ST-Mamba can reach optimal performance in a shorter time with less resource consumption compared to other Mamba-based CD networks.

D. Ablation Studies

In this section, a series of experiments will be conducted using the SYSU dataset to investigate the impact of various components of the proposed methodology on model performance, as shown in Table III.

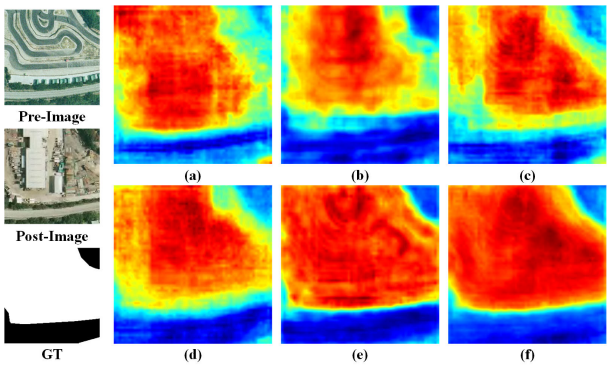


Fig. 10. Heatmap visualization of the feature maps for the ablation study results on the SYSU dataset illustrates the probability of change for each pixel. Areas closer to red indicate a higher likelihood of change, while areas closer to blue suggest a greater probability of nonchange. The red boxes highlight the parts that are specifically compared in this article. (a) Results of the baseline, (b) results of baseline + STFM, (c) results of baseline + ST-structure, (d) results of baseline + STSM, (e) results of baseline + ST-structure + STFM, and (f) results of baseline + STSM + STFM.

Five ablation experiments are conducted to validate the effectiveness of key modules in ST-Mamba. Among these, the baseline model represents a version of the ST-Mamba structure that excludes the primary enhancements, such as STSM and STFM. Specifically, the baseline model comprises four layers of encoders primarily constructed from MFEMs and four layers of decoders utilizing the second ST modeling approach, referred to as the concatenation modeling approach $S_c(\cdot)$, as described in Section III-E. It is important to note that we do not conduct ablation experiments specifically for MFEM, as its enhancements benefit the entire ST-Mamba structure. To demonstrate the intrinsic validity of the ST-Mamba structure, we introduced the ST-structure ablation term. Similar to STSM, the impact of pseudo-changes on the CD algorithm is mitigated in the decoder stage by employing pre-change and post-change feature maps to unify the background features. The key difference is that the STVCM approach and FPM are excluded, and only a straightforward summation method is utilized to apply the feature maps that unify the background features to the pre-change and post-change feature maps.

In this section, a series of experiments are conducted on the SYSU dataset to investigate the impact of each component of the proposed approach on the model performance, as illustrated in Table III.

Five ablation experiments were designed to validate the effectiveness of the key modules in ST-Mamba. The baseline is the basic Mamba structure. Its encoder consists of MFEM, and the decoder consists of VSS blocks. To demonstrate the intrinsic soundness of the ST-Mamba framework, we devised the ST-structure ablation term. It is located in the same place as the STSM and consists of the VSS block in the STSM with a residual transition module with only one convolution layer. Unlike other CD structures, where the two encoding branches do not interact during feature extraction, the ST structure adopts a simple structure that allows multitemporal features to interact during the encoding phase to create synergies, easily unifying the background features into a unified domain without filtering out the actually changing features of the other temporal phase.

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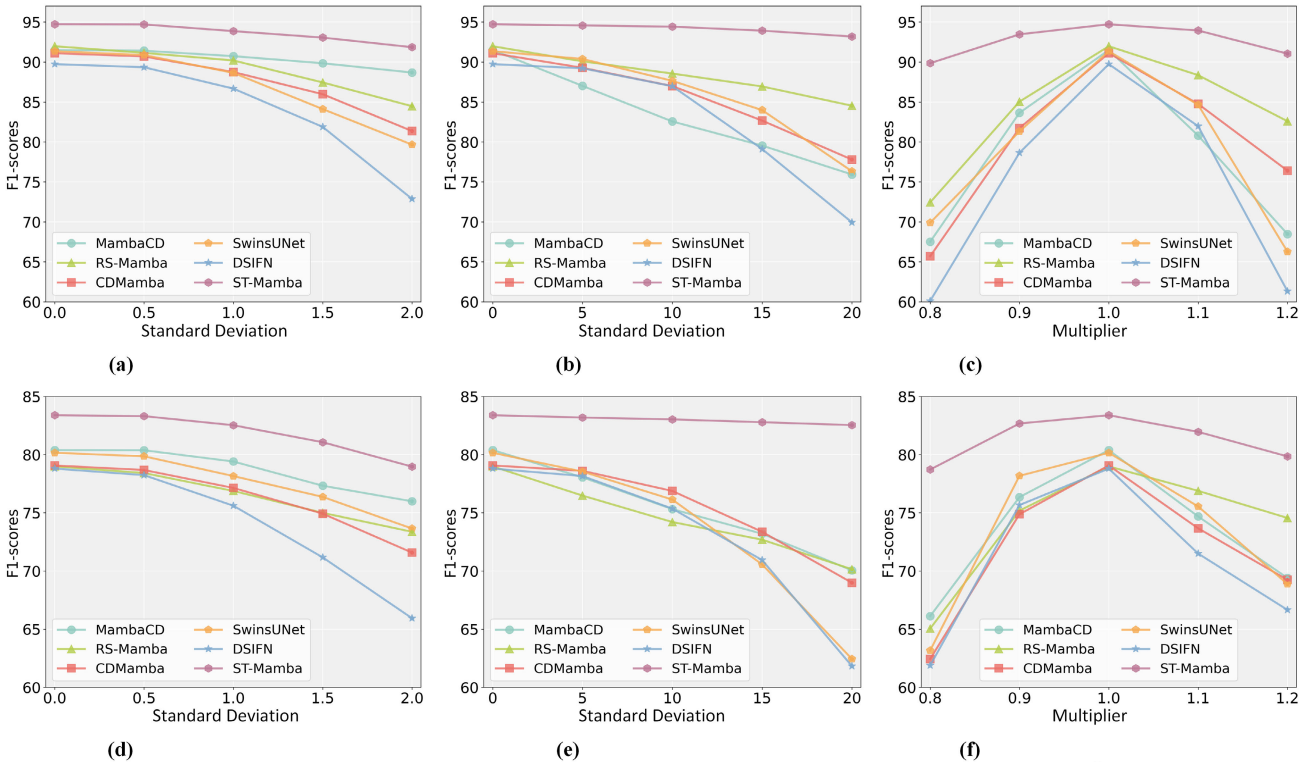


Fig. 11. Robustness of various models when tested against degraded images from the SYSU and WHU-CD datasets. (a) WHU GaussianBlur. (b) WHU GaussNoise. (c) WHU MultiplicativeNoice. (d) SYSC GaussianBlur. (e) SYSU GaussNoise. (f) SYSU MultiplicativeNoice.

From Table III, it is evident that the baseline model, a straightforward Mamba-based approach, outperforms more complex models such as MambaCD, RS-Mamba, and CDMamba on the SYSU dataset. Specifically, the baseline model surpasses the three existing Mamba-based baseline models in terms of $F1$-scores by 0.24%, 1.64%, and 1.56% further illustrating the effectiveness of the MFEM improvement. When STFM was incorporated into the baseline model, there was a significant enhancement in performance, with a 0.71% increase in the $F1$-score. The addition of either ST-structure or STSM alone resulted in similar improvements, with both $F1$-scores rising by 1%. This outcome highlights the effectiveness of ST-structure and supports the rationale behind the ST-Mamba framework. The advantages of STSM become particularly evident when it is used in conjunction with ST-structure or STFM. Specifically, when ST-structure and STFM were applied together to the baseline model, the $F1$-score improved by 1.94%. This represents a more pronounced enhancement in model performance compared to the application of ST-structure or STFM individually. Furthermore, when STSM and STFM were combined with the baseline model, there was an even more substantial improvement in performance compared to the ST-structure and STFM combination, with an increase in the $F1$-score of 0.82%. This result underscores the effectiveness of the STSM design.

The effects of the ablation experiment can be observed more intuitively through the images, as shown in Fig. 10. Compared to the baseline model, the change boundary is more distinct after incorporating the STFM, particularly in the upper right region. After incorporating ST-structure or STSM, the model detects the change region on the right side of the image with greater clarity. From the entire heat map, STSM slightly outperforms ST-structure in delineating the change region. Furthermore, when ST-structure and STFM are used in conjunction, it is evident that the model defines the change boundary with greater accuracy, which further demonstrates the effectiveness of the overall structure of ST-Mamba. When STSM and STFM are combined, the delineation of the change region is superior to that achieved with the combination of ST-structure and STFM, highlighting the effectiveness of the two modeling approaches of the STSM design and the FPM.

E. Robustness Against Degraded Input Data

Finally, we assessed the robustness of our method against degraded input data. During the inference phase, the data fed into the model were subjected to varying degrees of Gaussian blur, Gaussian noise, and multiplicative noise. Among these, Gaussian blur simulates image blurring caused by an improper focus during photography, Gaussian noise represents the disturbances that occur during image storage or transmission, and multiplicative noise reflects the conditions encountered when capturing images under excessively bright or too dark illumination.

The degradation of $F1$-scores due to data deterioration across various methods on the WHU-CD and SYSU datasets is shown in Fig. 11. As Gaussian blur is applied with an increasing standard deviation, the image becomes progressively blurrier. From Fig. 11(a) and (d), it is evident that the CNN-based DSIFN method exhibits the least robustness when subjected to increasing Gaussian blur, with degradation

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rates of 16.86% and 13.69% on the two datasets, respectively. In contrast, ST-Mamba and MambaCD showed the best and comparable robustness; at the highest standard deviation, their performance only decreased by approximately 2.8% and 4.4% on the two datasets. The performance degradation values of the other three methods are more similar, averaging around 9% and 7%. This indicates that ST-Mamba possesses superior robustness against image blurring.

When Gaussian noise is added, a higher standard deviation leads to more image noise. As illustrated in Fig. 11(b) and (e), ST-Mamba demonstrates remarkable resilience to Gaussian noise compared to other methods, with $F1$-scores decreasing by only 1.54% and 0.85%, respectively, at the maximum standard deviation. In contrast, the performance of the other methods shows significant declines, ranging from 7.44% to 19.79% on the WHU-CD dataset and from 8.84% to 17.72% on the SYSU dataset. This indicates that ST-Mamba exhibits excellent robustness against noise-like perturbations.

When multiplicative noise is added, a smaller noise parameter within the interval less than 1.0 results in a darker image, while a larger parameter within the interval greater than 1.0 produces a brighter image. As illustrated in Fig. 11(c) and (f), when the parameter decreases from 1.0 to 0.8, ST-Mamba experiences a reduction of 4.87% and 4.67% on the two datasets, respectively, under darker conditions. When the parameter increases from 1.0 to 1.2, ST-Mamba shows a decrease of 3.68% and 3.54% on the two datasets in the same darker scenario. In comparison, other methods are more adversely affected by multiplicative noise. On the WHU-CD dataset, the $F1$-scores of all methods dropped by more than 20% when the noise parameter was set to 0.8, and similarly by more than 20% when the parameter was set to 1.2, with the exception of RS-Mamba, which only decreased by 9.38%. The situation on the SYSU dataset mirrors that of the WHU-CD dataset, with reductions of approximately 15% under darker images and around 10% under lighter images, again with RS-Mamba showing a smaller decrease of only 4.43%. This clearly demonstrates that ST-Mamba exhibits superior robustness in response to variations in image brightness.

V. Conclusion

This article introduced a novel CD model, ST-Mamba, which employs an innovative Mamba-based framework. Specifically, after the initial extraction of image features by MFEM, the feature maps are input into the STSM synergistically utilizes multitemporal image features to unify the background information of images into a unified domain. Subsequently, the STSM leverages these unified background features to guide multiscale feature fusion and channel aggregation. Ultimately, the final CD results are produced through a stepwise decoding process. Our extensive experiments on five widely used datasets demonstrated that ST-Mamba outperformed existing SOTA methods and effectively addressed the issue of other Mamba-based models being highly susceptible to pseudo-changes when detecting change regions. Furthermore, our proposed ST-Mamba exhibited reduced time overhead and achieved optimal results in a shorter duration. Moreover, ST-Mamba showed the highest level of robustness when confronted with various classes of realistic data degradation phenomena. Nevertheless, when faced with the simultaneous occurrence of multiple complex scenarios—such as variations in scenes combined with the presence of small, occluded objects in images, irrelevant objects causing occlusions alongside significant differences in imaging conditions, or substantial disparities in illumination coexisting with low image resolution—ST-Mamba struggles to achieve fully accurate CD. In future work, we aim to enhance the robustness of CD networks to accommodate a broader range of complex scenarios, thereby further advancing remote sensing technology.

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